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New paleomagnetic constraints on the extrusion of Indochina: Late Cretaceous results from the Song Da terrane, northern Vietnam

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Abstract

Samples were collected for paleomagnetic investigations from the Upper Cretaceous Yen Chau Formation in the Song Da terrane (21.7°N, 103.9°E) bounded by the Ailao Shan–Red River fault system and Song Ma fault, in an attempt to examine extrusion tectonics of East Asia. Primary nature of the high-temperature (600–690 °C) component of magnetization from 13 sites is supported by a positive fold test. The tectonic corrected data provides the characteristic Late Cretaceous paleomagnetic direction for the Song Da terrane (D=6.4°, I=32.0° with α_{95}=8.5°, N=13 sites), corresponding to a paleopole lying at 82.9°N, 220.7°E (A_{95}=6.9°). Comparison with coeval paleomagnetic poles for the neighboring tectonic blocks indicates no latitudinal translation of the Son Da terrane with respect to the South China Block, whereas the Shan-Thai Block and the southern part of the Indochina Block revealed a southward displacement of 10.5±9.5° in latitude with respect to the Song Da terrane. We conclude that the southeastern segment of the Ailao Shan–Red River fault system to the east of the Dien Bien Phu fault is not a demarcation of the extruded Indochina Peninsula. The Indochina Block extruded probably along some faults between the Song Da terrane and the Khorat Plateau.
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1. Introduction

The Indo-Asian collision and the subsequent northward penetration of India have resulted in the extrusion tectonics in the eastern part of Asian continent (e.g. [1–4]). The Indochina Peninsula is composed of the Indochina Block, Shan-Thai Block,

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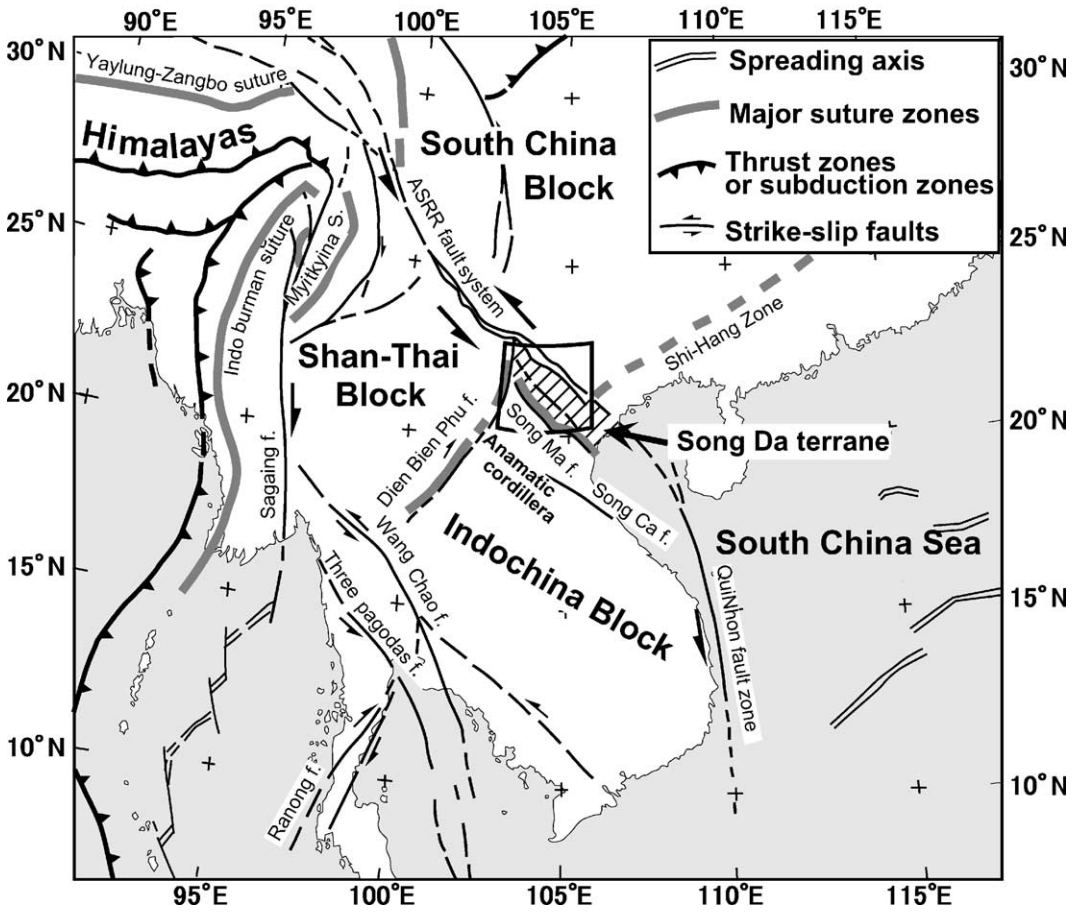


Fig. 1. Structural sketch map of Southeast Asia (modified from Leloup et al. [5]). Enlargement of the bracketed area is shown in Fig. 2a. The arrows aside of faults correspond to Tertiary shear senses prior to 16 Ma.

and the southern portion of the South China Block (SCB) (Fig. 1). Recent paleomagnetic studies provide quantitative amounts of Indochina and Shan-Thai blocks extrusion. Cretaceous red beds from Yunlong in the northern part of the Shan-Thai Block and the Khorat plateau in the southern part of the Indochina Block show a southward displacement of around 10° in latitude together with clockwise rotation around vertical axis of more than 14° with respect to the SCB [6–9], suggesting strongly that the Indochina Peninsula was subjected to a southward displacement exceeding 1000 km after the Cretaceous.

The amalgamation of the Indochina Block with the SCB has continued until the Late Triassic on the basis of geologic and paleontological studies [10–14]. Within the region of the boundary between the Indochina and South China blocks, there are several major NW–SE trending faults: e.g. the Ailao Shan–Red River (ASRR) fault system (including the Song Chay and Red River faults), the Song Da, Song Ma and Song Ca faults (Figs. 1 and 2a). Some authors considered that the left-lateral movement along the ASRR fault system played an important role in the extrusion model for the Indochina Peninsula (e.g. [1,15,16]), although it is currently active as right-slip [17,18].

The Song Da terrane is located between the ASRR fault system and Song Ma fault. The nature of the Song Da terrane displacement offers key constraints on the accommodation of the southward extrusion of the Indochina Peninsula. In this study, we present paleomagnetic results from the Song Da terrane. We focus on the red beds of the Upper Cretaceous Yen

Chau Formation because they could have recorded aspects of the deformation of Indochina later than 45 Ma, the time at which India collided with Asia [19].

2. Geology and sampling

Late Cretaceous to Early Cenozoic continental sedimentary rocks are widely distributed in the Song Da terrane (Fig. 2a) [20,21]. The Upper Cretaceous Yen Chau Formation overlies unconformably upon Paleozoic and Triassic sediments, and is unconformably overlain by the Paleogene Pu Tra Formation [20]. The Yen Chau Formation is 850 to 1300 m thick at the type locality (21°03′ N, 104°18′ E) that is subdivided into lower and upper members. The lower one is characterized by red colored conglomerate to coarse sandstones, and the upper one by red colored coarse sandstones to siltstones. The occurrence of *Unio* sp. aff. *U. grabaui*, *Ulmus* sp., *Zelkova* sp., *Srundo* sp., *Gyraulus* sp., *Betula* cf. *cuspidens*, *Laurus* cf. *similis* assigns a Late Cretaceous age to this formation [20].

Paleomagnetic samples were collected from the upper member of the Yen Chau Formation in the Yen Chau locality (21.0°N, 104.4°E, Fig. 2a). Sampling sites are distributed along a road over about 50 km from northwest to southeast (Fig. 2b). In this locality, the Yen Chau Formation forms a synclinal structure with a NW–SE-fold axis, according to the cross section on the basis of detailed geological survey [22]. The stereo-plot of stratification shows that each site tilted around a single horizontal fold axis with NW–SE trend (Fig. 2c). Eighty samples were taken at 10 sites in both limbs of synclinal structure for the fold test.

Samples were also collected from the upper member of the Yen Chau Formation in the Lai Chau locality (22.3°N, 103.4°E). This locality lies about 180 km to the northwest of the Yen Chau locality (Fig. 2a) and covers a smaller area than the Yen Chau locality (Fig. 2d). The geological structure is characterised by NW–SE axis folding on the basis of the stereo-plot of stratification (Fig. 2e). Fifty-two samples were taken at six sites in both limbs in this locality.

Six to ten hand samples were distributed within a distance up to 30 m at each site. The location of each sampling site was determined by a portable GPS. Samples were oriented by a magnetic compass. The present geomagnetic direction at each site is calculated using the International Geomagnetic Reference Field, 2000 [23].

3. Paleomagnetic procedure and results

One or more specimens, 25 mm in diameter and 22 mm in height, were prepared for each hand sample in the laboratory. Natural remanent magnetization (NRM) of all specimens was measured with a 2G Enterprises cryogenic magnetometer. Stepwise thermal demagnetization up to 690 °C was performed for each specimen in order to isolate remanent magnetic components. Thermal treatments were carried out with a Natsuhara TDS-1 thermal demagnetizer; residual magnetic field in this furnace during the cooling cycle was less than 5 nT. Bulk susceptibility was also measured at each demagnetization step with a Barington MS2 susceptibility system to monitor any alteration of magnetic minerals. Stepwise alternating field demagnetization (AFD) up to 100 mT was also carried out for at least one specimen per site with a Natsuhara DEM-8601 AF demagnetizer. In order to identify ferromagnetic minerals based on magnetic coercive force, progressive acquisition of isothermal remanent magnetization (IRM) up to 1.0 T was performed for one specimen per site with a 2G pulse magnetizer.

The magnetic behavior of each specimen during demagnetization was plotted on orthogonal vector diagrams [24] as well as on equal area projection to assess component structure. Directions of the remanent magnetization components were determined using principal component analysis with a maximum angle of deviation less than 10.0° [25]. Site mean directions were calculated for sites with at least three samples using Fisherian statistics [26]. Locality mean directions were based on the expected directions at 21.0°N/104.4°E for Yen Chau locality and at 22.3°N/103.4°E for Lai Chau locality. Paleomagnetic pole positions and their uncertainty were calculated using the method described by Butler [27].

NRM intensity ranges between 0.58 and 13.0 mA/m. Stepwise thermal cleaning of samples revealed multicomponent NRM; generally, two components appear(Fig. 3a–e). A low-temperature component (LTC) is erased between 120 and 570 °C. After removal of this LTC, remanent magnetization decays

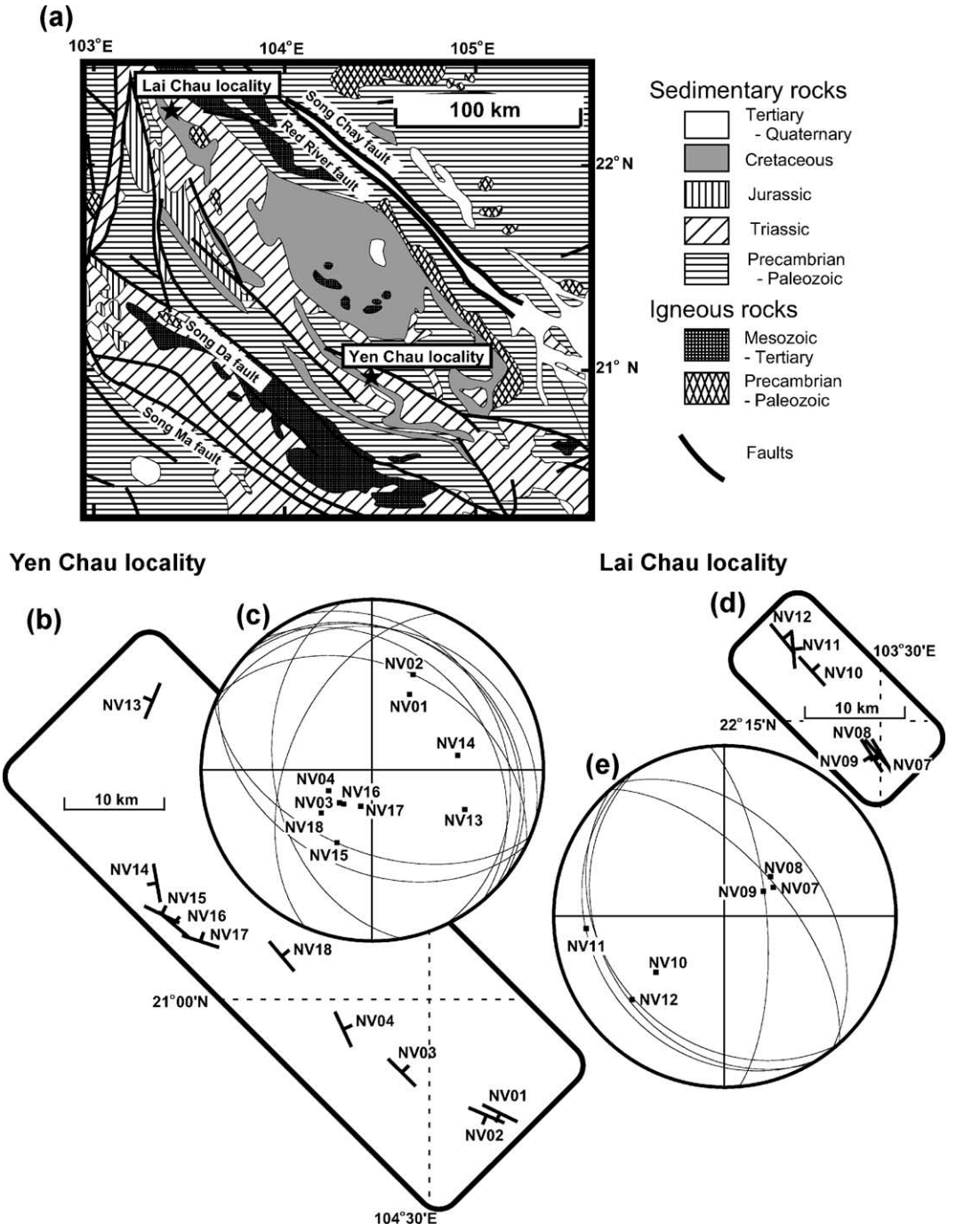


Fig. 2. (a) Simplified geological map of studied area (modified from the geological map published by Tingdong et al. [21]). Locations of the studied localities are marked by stars. (b, d) Distributions of sampling sites. (c, e) Structural data (Schmidt diagrams, lower hemisphere). Great circles, bedding planes; squares, pole of bedding planes.

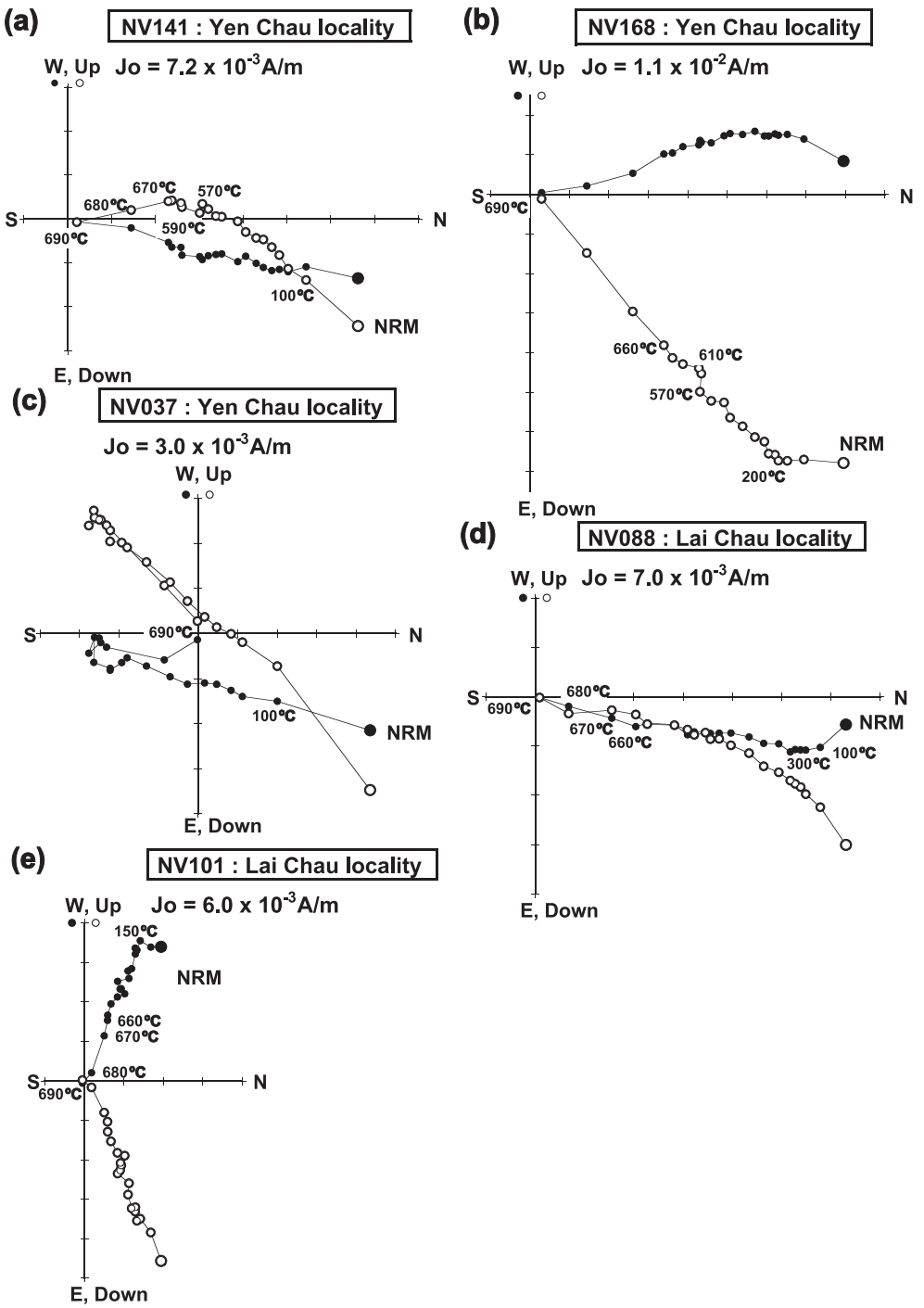


Fig. 3. Orthogonal projections of magnetization vector end points through stepwise thermal demagnetizations. Solid (open) symbols are for horizontal (vertical) projection. All diagrams are in geographic coordinate.

linearly to the origin allowing us to resolve a high-temperature component (HTC) that is unblocked between 660 and 690 °C.

AFD up to 100 mT is ineffective. Indeed, this technique either destroyed a small part of, or completely failed to reduce, the intensities of NRMs, implying that high coercive force minerals mainly carry NRMs (Fig. 4a and c). The IRM acquisition curves (Fig. 4b and e) reveal that no saturation of intensities is achieved after applying 1.0 T, suggesting the predominance of high coercive force minerals. The unblocking temperature of the HTCs and these two results indicate that hematite carries the HTCs of the Yen Chau Formation [28].

3.1. Yen Chau locality

LTCs with normal polarity were isolated from 10 sites. Their locality mean in geographic coordinate is D=359.2°, I=36.7° (k=15.8, α_{95}=12.5°), which is parallel at 95% confidence level to the geocentric axial dipole (GAD) field direction at this locality (D=0.0°, I=37.5°). Moreover, the fold test [29] is negative at 95% confidence level (k_g/k_s=2.43>F_c). These LTC

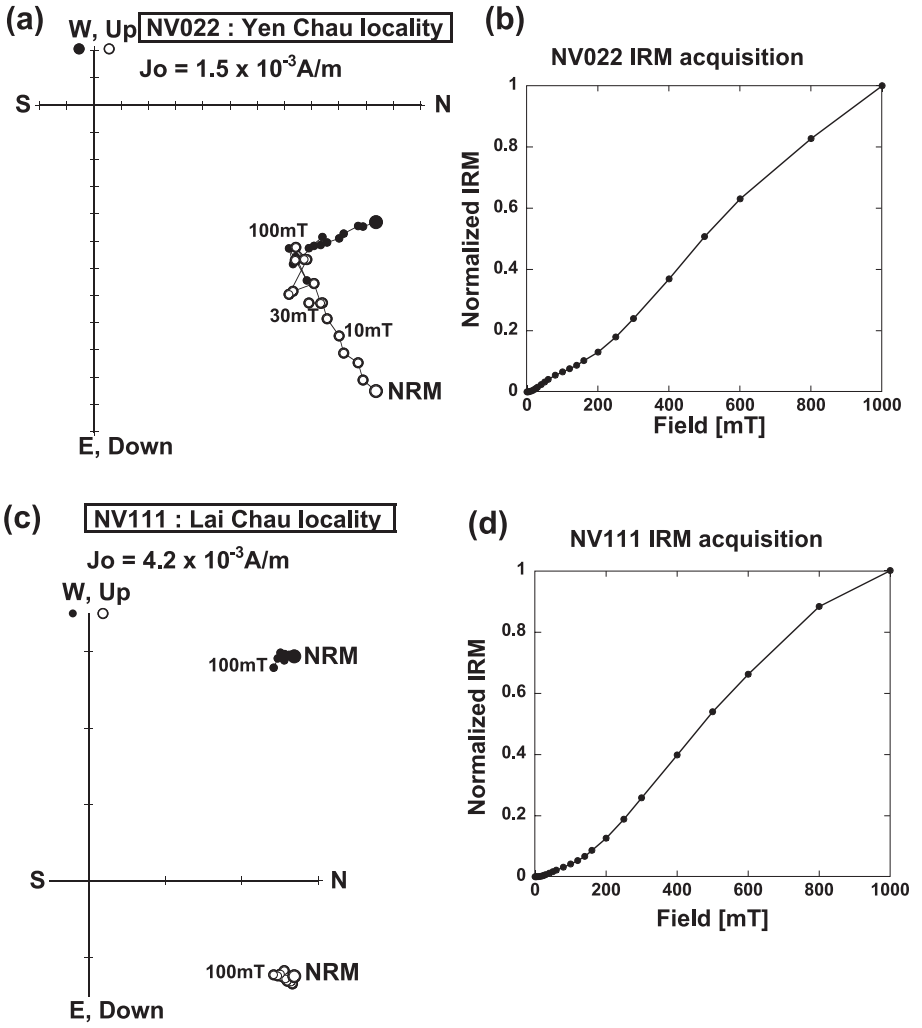


Fig. 4. (a, c) Orthogonal projections of magnetization vector end points through stepwise alternating field demagnetization in geographic coordinate. Solid (open) symbols are for horizontal (vertical) projection. (b, d) Isothermal remanent magnetization acquisition curves normalized by the intensities at 1000 mT.

directions were thus interpreted as secondary magnetizations acquired during the Bruhnes chron.

HTCs were isolated from 10 sites (Fig. 5a). Seven sites have normal polarity and three sites reversed polarity. We calculate a locality mean of HTCs from the 8 of 10 sites after flipping the reversed polarity directions into normal polarity. Two sites were discarded from this mean calculation because of a large dispersion (site NV04, α_{95}=33.5°) and a small number of samples (site NV13, n=3). This locality mean on the basis of eight sites is D=0.3°, I=26.1° (k=7.2, α_{95}=22.1°) before, and D=3.2°, I=26.7°

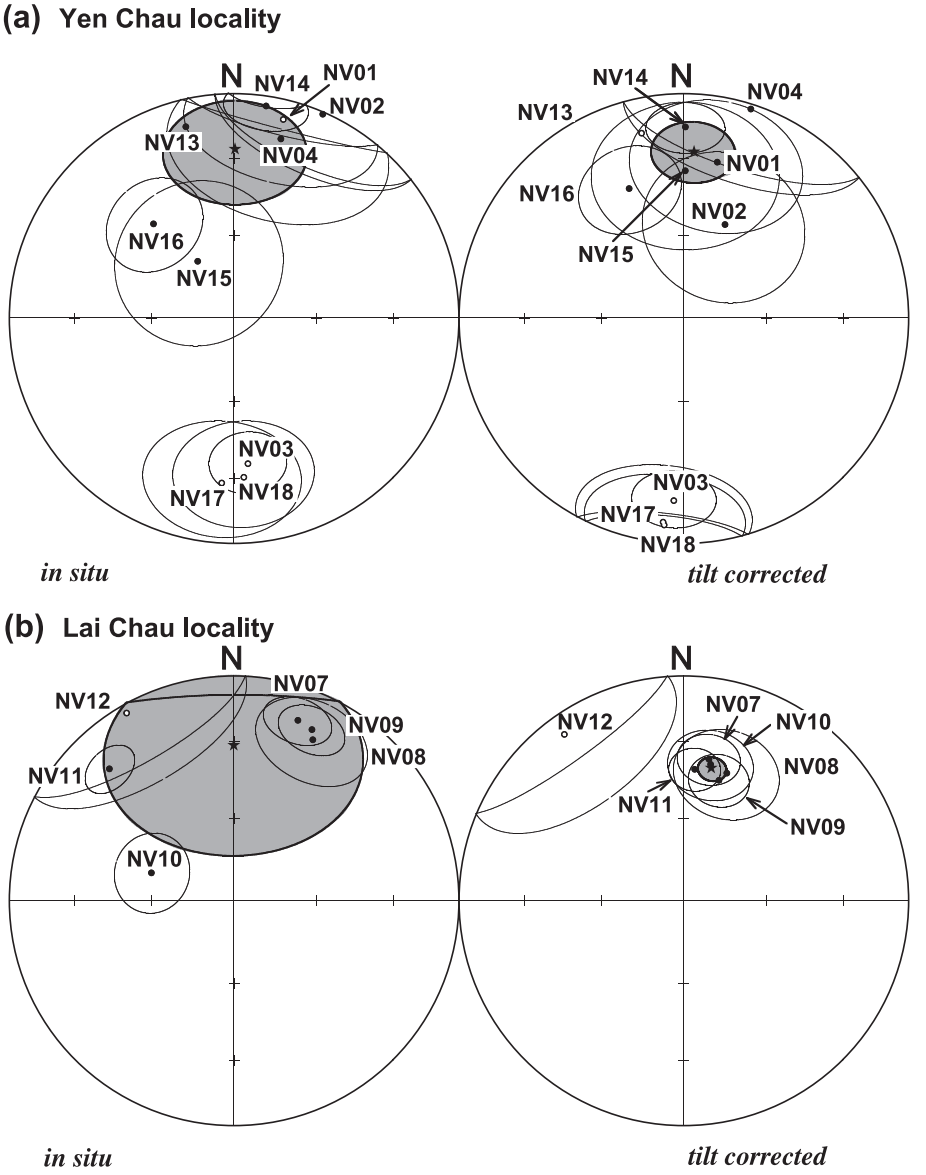


Fig. 5. Equal area projections of site mean directions of the high temperature components at Yen Chau locality (a) and Lai Chau locality (b). Left (right) diagram is in geographic (stratigraphic) coordinates (Table 1). Solid (open) symbols show downward (upward) directions. Locality mean directions and their 95% confidence limits are shown as stars and shaded circles (Table 1). Sites NV04, 12, 13 were discarded from calculation of locality means (see in the text).

(k=19.4, α_{95}=12.9°) after tilt correction. Alternatively, we also calculate a selected mean direction from better determined three sites (sites NV03, 14 and 16) with α_{95}<20°. This selected mean on the basis of three sites is D=350.4°, I=30.1° (k=7.7, α_{95}=47.7°) before and D=354.6°, I=25.1° (k=23.3, α_{95}=26.1°) after tilt correction. Because two mean directions are indistinguishable and the locality mean characterizes wider area than the selected mean, we recognize the locality mean direction on the basis of eight sites as the characteristic paleomagnetic direction of the Yen Chau locality.

The locality mean direction on the basis of eight sites passes the positive fold test at 95% confidence level (k_s/k_g=19.4/7.2>F_c) [29]. The direction–correction (DC) tilt test [30] gives the optimal concentration at 79±38% tilt correction, which is indistinguishable from 100% tilt correction. These results indicate the pre-folding origin. The presence of normal and reversed polarity, with a positive reversal test [31] in classification "I" (the angular distance γ=22.1° is smaller than the critical angle γ_c=22.9°) after tilt correction, suggests that geomagnetic secular variation has been averaged sufficiently.

Table 1
High-temperature components of the Upper Cretaceous Yen Chau Formation from the Song Da terrane

Site	Location		Strike (°)	Dip (°)	n/N	In situ		Tilt corrected		k	α_{95} (°)
	Lat (°N)	Long (°E)				D (°)	I (°)	D (°)	I (°)		
Yen Chau locality											
NV01	20°53′ 34.6″	104°34′ 11.0″	116	41	3/8	13.9	−10.1	12.1	29.9	21.1	27.5
NV02	20°53′ 32.1″	104°33′ 14.4″	113	51	5/9	23.5	1.5	23.7	52.5	8.2	28.4
NV03	20°55′ 55.9″	104°28′ 25.9″	315	22	5/8	174.5	−35.6	183.1	−20.3	38.6	12.5
NV04*	20°58′ 18.3″	104°24′ 56.8″	335	23	9/9	14.6	18.9	17.8	2.9	3.3	33.5
NV13**	21°16′ 18.0″	104°13′ 41.7″	203	49	2/6	345.9	13.5	347.2	−16.9	–	–
NV14	21°06′ 23.5″	104°14′ 02.0″	170	42	7/7	8.6	5.4	0.5	16.3	29.2	11.4
NV15	21°04′ 41.4″	104°14′ 21.4″	296	39	4/8	327.5	65.5	0.6	34.9	10.2	30.3
NV16	21°04′ 06.5″	104°14′ 58.6″	310	21	4/8	319.6	44.2	337.2	37.6	29.7	17.1
NV17	21°03′ 07.4″	104°16′ 37.2″	288	18	6/8	184.3	−27.7	185.8	−10.2	8.4	24.6
NV18	21°02′ 17.4″	104°21′ 14.6″	320	32	7/9	176.5	−30.2	185.4	−8.9	8.3	22.2
Locality mean	21.0°	104.4°			8/10	0.3	26.1	–	–	7.2	22.1
						–	–	3.2	26.7	19.4	12.9
								Lat (°)	Long (°)	K	A_{95} (°)
Corresponding pole					8/10			83.2	255.6	27.2	10.8
Lai Chau locality											
NV07	22°13′ 06.9″	103°27′ 02.3″	149	27	7/7	19.5	16.3	10.1	36.5	35.8	10.2
NV08	22°13′ 14.5″	103°26′ 49.0″	139	29	8/8	24.5	17.8	16.3	43.7	30.7	10.2
NV09	22°13′ 08.6″	103°26′ 29.9″	147	22	8/10	26.1	21.7	18.7	40.1	10.9	17.6
NV10	22°17′ 52.3″	103°22′ 59.5″	318	62	10/10	288.6	58.3	11.2	38.4	12.9	14.0
NV11	22°19′ 03.6″	103°21′ 54.4″	355	70	9/9	316.6	20.7	4.7	41.2	37.9	8.5
NV12*	22°19′ 36.7″	103°21′ 13.7″	321	43	8/8	330.2	−4.7	324.3	−10.2	3.8	33.0
Locality mean	22.3°	103.4°			5/6	359.7	32.1	–	–	4.3	41.7
						–	–	12.2	40.1	263.9	4.7
								Lat (°)	Long (°)	K	A_{95} (°)
Corresponding pole					5/6			78.7	188.0	222.8	5.1
Formation mean	21.7°	103.9°			13/15	0.1	28.3	–	–	6.3	18.0
						–	–	6.4	32.0	24.7	8.5
								Lat (°)	Long (°)	K	A_{95} (°)
Characteristic Late Cretaceous paleomagnetic pole					13/15			82.9	220.7	36.6	6.9

Lat and Long=latitude and longitude; n/N=numbers of samples or sites used for statistics/total numbers of collected samples or sites; D and I=declination and inclination; k and α_{95} (K and A_{95})=the Fisherian precision parameter [26] and half angle of the 95% confidence cone of mean direction (pole position). Data marked with * and ** are discarded from the calculation due to the large dispersion and due to small number of samples, respectively (see text).

3.2. Lai Chau locality

The locality mean of the LTCs in geographic coordinates computed for six sites is D=358.9°, I=36.7° (k_g=103.5, α_{95}=6.6°). These directions reveal a fold test that is negative at 99% confidence level (k_g/k_s=16.7>F_c) [29]. Therefore, as for the Yen Chau locality, the LTCs isolated from this locality are interpreted as secondary and acquired in the Bruhnes chron.

Normal polarity HTCs were isolated from six sites (Fig. 5b and Table 1). Site NV12 was excluded from the computation for the locality mean due to its large dispersion (k=3.8) (Fig. 5b). The locality mean HTC based on remaining five sites is D=359.7°, I=32.1° (k=4.3, α_{95}=41.7°) before, and D=12.2°, I=40.1° (k=263.9, α_{95}=4.7°) after tilt correction (Table 1). These directions pass a positive fold test at 99% confidence level (k_s/k_g=61.4<F_c) [29]. The correlation test [32] also reveals a positive result at 95% confidence level (ξ_g=3.038 in situ, ξ_s=0.597 after tilt correction and critical value ξ_c=2.609). The DC tilt test [30] gives the optimal concentration at 101±14% tilt correction, which is indistinguishable from 100% tilt correction. These results taken together strongly indicate a pre-folding origin of HTCs.

4. Discussion

4.1. Timing of magnetization

The HTCs resolved in the continental red beds in both Yen Chau and Lai Chau localities are interpreted as primary magnetizations of the Upper Cretaceous Yen Chau Formation. The pre-folding nature of the HTCs is indicated by positive fold tests and the presence of both polarities. Because Cretaceous red beds in the Song Da terrane are deformed by early Tertiary shortening [22], we conclude that the HTCs of the Yen Chau and Lai Chau localities were acquired in the Late Cretaceous.

4.2. Characteristic Late Cretaceous paleomagnetic pole for the Song Da terrane

Similar paleomagnetic pole positions of the Late Cretaceous have been obtained from the Yen Chau and Lai Chau localities (83.2°N, 255.6°E with A_{95}=10.8° and 78.7°N, 188.0°E with A_{95}=5.1°, respectively), although two localities are separated by ~180 km (Fig. 2a and Table 1). They are compared with coeval paleomagnetic poles for the SCB (Fig. 6a and Table 2). The Late Cretaceous paleomagnetic pole for the whole SCB (74.3°N, 202.4°E, with A_{95}=9.9°) [33] is located close to our poles: The angular distance from the pole for the whole SCB to the poles for Yen Chau and Lai Chau localities are 12.8±14.7° and 5.5±11.0°, respectively. Recently, a Cretaceous paleomagnetic pole for the stable body of SCB (80.0°N, 206.7°E with A_{95}=2.5°) has been reported [34], which is located between our two poles (Fig. 6a). Comparison with it shows that the angular distances are 7.5±11.1° and 3.7±5.3° from the pole for the stable body of SCB to the pole for the Yen Chau and Lai Chau localities, respectively. These evaluations indicate that two localities in the Song Da terrane were subjected to little relative movement with respect to the SCB after the Late Cretaceous.

The pole positions for the Yen Chau and Lai Chau localities statistically agree with each other 95% confidence level: The angular distance between two poles is 10.7±11.8° at 95% confidence level (Fig. 6a). This good agreement between two poles indicates insignificantly large intra-terrane deformation between two localities. Assuming that the Song Da terrane has behaved as a single rigid tectonic unit after the Late Cretaceous as a first-order approximation, we evaluate the characteristic Late Cretaceous paleomagnetic pole for the Song Da terrane on the basis of 13 sites from both localities as situated at 82.9°N, 220.7°E with A_{95}=6.9° (Table 1, Fig. 6a), which provides a characteristic direction at the reference point in this terrane (21.7°N, 103.9°E) as D=4.9°, I=32.5° with α_{95}=6.9°.

This characteristic Late Cretaceous pole position for the Song Da terrane is then compared with those for the SCB (Fig. 6a). Our Late Cretaceous pole for the Song Da terrane also agree with the poles for the whole SCB [33] and for the stable body of SCB [34] at 95% confidence level: The angular distances are 9.2±12.1° and 3.6±7.3°, respectively. Because of excellent agreement of pole positions, we conclude that no relative movement occurred between the Song Da terrane and the SCB after the Late Cretaceous.

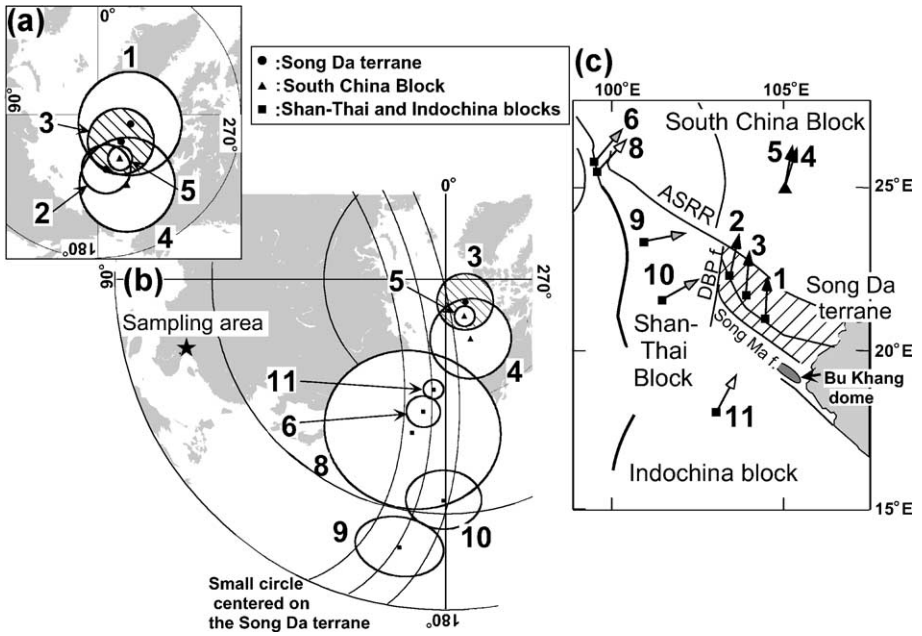


Fig. 6. Cretaceous paleomagnetic results from the Indochina Peninsula and the South China Block. Number beside each pole/direction corresponds to Table 2. (a, b) Equal area projection of the Cretaceous paleomagnetic poles with 95% confidence circle for the Song Da terrane (circle), the South China Block (triangle), the Shan-Thai and Indochina blocks (square). The pole with hatched confidence circle is the characteristic Late Cretaceous pole for the Song Da terrane. Radius of the small circle centered on our sampling area (21.7°N/103.9°E) is 61.1±6.4° (2σ), calculated as an arithmetic mean of the colatitudes from the sampling area for poles of 6, 8–11. (c) Declinations after flipping into the normal polarities. Black, gray and white arrows show the Late, middle and Early Cretaceous results, respectively. ASRR, Ailao Shan–Red River fault system; DBP f., Dien Bien Phu fault.

4.3. Tectonically stable feature of the Song Da terrane

The Late Cretaceous paleomagnetic pole position for the Song Da terrane is compared with coeval paleomagnetic poles for the Shan-Thai Block and the southern part of the Indochina Block (Fig. 6b and Table 2). The Song Da pole is located at far-sided position from all five Cretaceous poles for the Shan-Thai Indochina blocks (Fig. 6b) [6–8,35]. We calculate a small circle centered on the Song Da terrane (21.7°N, 103.9°E) which is passing these five poles. The radius of this small circle and its uncertainty (±2σ) is 61.1±6.4°, which are evaluated from the arithmetic mean and standard deviation (σ) of colatitudes from the Song Da terrane to five poles, respectively. The angular distance from this small circle to the Song Da pole is 10.5° and its uncertainty (Δd) is 9.5°, where Δd is given as $[(A_{95})^2+(6.4^\circ)^2]^{1/2}$ and A_{95} is the radius of 95% confidence circle of the Song Da pole. We conclude that the characteristic Late Cretaceous pole for the Song Da terrane occupies significantly far-sided position from five poles by 10.5±9.5°. This implies a southward displacement of the Shan-Thai and Indochina blocks is 1200±1000 km with respect to the Song Da terrane. The distribution of five Cretaceous poles for the Shan-Thai and Indochina blocks also indicates large clockwise rotation of these blocks with respect to the Song Da terrane by more than 22° (Fig. 6b). These results indicate that the post-Cretaceous tectonic behavior of the Song Da terrane is quite different from that of the Shan-Thai Block and the southern part of the Indochina Block.

On the basis of the distribution of the Cretaceous paleomagnetic poles for the Indochina Peninsula and South China, we conclude that the Song Da terrane behaved as a part of the SCB, when the Shan-Thai Block and the southern part of the Indochina Block experienced a southward displacement accompanied with a clockwise rotation later than the Indo-Asian collision.

Table 2
Cretaceous paleomagnetic results from the Indochina Peninsula

No. area	Location		Age	N	Observed direction			Pole position			Clockwise rotation (°)	Southward displacement (°)	Reference
	Lat (°N)	Long (°E)			D (°)	I (°)	α_{95} (°)	Lat (°N)	Long (°E)	A_{95} (°)			
Song Da terrane													
(1) Yen Chau	21.0	104.4	K3	8	3.2	26.7	12.9	83.2	255.6	10.8	–±–	–±–	this study
(2) Lai Chau	22.3	103.4	K3	5	12.2	40.1	4.7	78.7	188.0	5.1	–±–	–±–	this study
(3) Mean	21.7	103.9	K3	13	4.9	32.5	11.4	82.9	220.7	6.9	–±–	–±–	this study
South China Block													
(4) Whole			K2–3	5				74.3	202.4	9.9	–±–	–±–	[33]
(5) Stable body			K1–3	5				80.0	206.7	2.5	–±–	–±–	[34]
Shan-Thai Block													
(6) Yunlong	25.8	99.4	K2	20	40.2	49.9	3.9	54.6	171.3	4.4	33.5±9.0	9.2±8.2	[8]
(7) Yunlong	25.8	99.4	K2	29				56.7	180.1	4.0	31.0±8.8	9.8±8.0	[9]
(8) Yongping	25.5	99.5	K1	12	42.0	51.1	15.7	50.9	167.3	20.6	37.8±26.0	11.6±21.7	[6]
(9) Jinggu	23.4	100.9	K2	8	79.4	43.3	9.1	18.9	170.0	8.9	73.0±12.3	6.3±11.3	[35]
(10) Mengla	21.6	101.4	K2	10	60.8	37.8	7.6	33.7	179.3	8.2	54.5±11.4	3.5±10.7	[35]
Indochina Block													
(11) Khorat	16.5	103.0	K1	10	28.1	40.5	2.4	62.7	173.3	2.4	21.7±7.6	10.5±7.3	[7]

Lat and Long=latitude and longitude; N=numbers of sites used in the mean calculations; K1, K2 and K3=Early, middle and Late Cretaceous, respectively; D and I=declination and inclination; α_{95} and A_{95}=half angles of 95% confidence cone of paleomagnetic direction and paleomagnetic pole position. Clockwise rotation and southward displacement are evaluated as the relative motion of each studied area with respect to the Song Da terrane by the method described by Butler [27].

4.4. Tectonic implications

Our paleomagnetic conclusion is supported by geological evidence. The Indochina and South China blocks were amalgamated by the Late Triassic [10–13], and the suture between two blocks is located along the Song Ma fault, the southern margin of the Song Da terrane [14,36] (Figs. 1 and 6c). The presence of mafic and ultramafic rocks and greenschists, including mélange, along the Song Ma fault marks the suture between the Indochina and South China blocks [37]. High-grade metamorphic rocks along the Song Ma fault are dated at about 245 Ma by $^{40}Ar/^{39}Ar$ plateau cooling ages, implying that the final amalgamation between the Indochina and South China blocks took place along the Song Ma fault in the Early Triassic [38]. Therefore, the Song Da terrane was geologically the southern portion of the SCB before the Indo-Asian collision.

Because the extrusion model for the Indochina Peninsula requires sinistral motion along a prominent strike-slip fault, the ASRR fault system has been suggested as the most plausible tectonic boundary between the extruded terranes and the stable SCB (e.g. [1,16,39]). The left-lateral shear along the ASRR fault system has been postulated on the basis of thermochronological studies for exhumed high-grade mylonitic gneisses (e.g. [5,40]). However, little direct evidence for the "large" left-lateral movement are observed in a fault zone to the east of the Dien Bien Phu Fault (Table 2 and Fig. 34 of Leloup et al. [16]), although left-lateral shear is detected from the Day Nui Con Voi massif (e.g. [5]). Geophysical research reported from the offshore area of northern Vietnam has revealed that the sinistral offset of the Red River Fault in the Tonkin Gulf may not have exceeded a few tens of kilometers on the basis of seismic profile calibrated by deep wells [41], indicating the absence of the "large" left-lateral movement to the east of the Dien Bien Phu Fault. These geological and geophysical data indicate no significantly "large" relative movement between the Song Da terrane and the SCB along the ASRR fault system.

Nonetheless, Cretaceous paleomagnetic results require a southward displacement of the Shan-Thai Block and the southern part of the Indochina Block

with respect to the Song Da terrane and SCB. Because geologic markers indicating the "large" left-lateral displacement (offsets of ophiolites, Mesozoic red bed basins, Triassic arc volcanic rocks, Cretaceous granites and Eocene high-K igneous rocks) are distributed outside of the Song Da terrane (e.g. [16]) and the paleomagnetic data revealed southward displacement of the Khorat Plateau [7], the extrusion of Indochina occurred along a fault (or faults) located between the Song Da terrane and the Khorat Plateau. Exhumation of the Bu Khang extensional dome south of the Song Ma fault [42,43] seems to have occurred under the condition of a transtensional stress field of a transtension with NE–SW extension (Fig. 6c). Therefore, we alternatively propose that the Song Ma fault, to the south of the Song Da terrane, played an important role during the extrusion of Indochina. More geological investigations and paleomagnetic data in northern Vietnam and Lao are required to understand the extrusion tectonics in the Indochina Peninsula due to the Indo-Asian collision.

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